Docking in Drug Discovery for Glucagon-Like Peptide-1 Receptor

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I. Introduction

Glucagon-Like Peptide-1 Receptor (GLP-1R) is a regulator in insulin release and the pk13/Akt pathway[1]. GLP-1R is activated by GLP-1 when it binds [1]. The structural conformation of GLP-1R changes when bound to GLP-1, and it exists "in thermal equilibrium among different states by sampling multiple conformations"[2]. Understanding the conformations which GLP-1R can adopt is critical to the identification of agonists. Molecular docking is the "identification of the low-energy binding modes of a small molecule, or ligand within the active site of a macromolecule, or receptor, whose structure is known"[3]. This article provides an analysis of potential agonists from docking for GLP-1R utilizing the visualization software, PyMOL 3.1 by Schrödinger for Educational–Use-Only [4]. All files can be found at https://github.com/jmmoses99/Structural_Bio.git

II. METHODS

To perform docking, a protein of interest is required, (structure can be found in PDB or homology modeling can be used), a docking site on the protein of interest is needed, and a ligand database (i.e. Zinc) must be used [3]. Docking depends on the search algorithm and scoring function to provide the top results for potential agonists [3]. The PyMol session with docking results for GLP-1R was provided. The file was opened in PyMol [4]. GLP-1R was labeled as an object in the session under the name rec. A ligand was found in the binding pocket of GLP-1R and labeled as an object in the session under the name xtal-lig. This indicated the GLP-1R structure was in an activated state. To activate GLP-1R, a competitive agonist to the ligand, xtal-lig would be needed. There were 1000 potential agonists generated by docking in the PyMol file under the name glp_poses. Initially, the glp_poses were sorted through using the forward and reverse arrows in PyMol. However, this is a highly inefficient method to evaluate the glp_poses. PyMol includes a command line that utilizes the programming language, Python. This command line was critical in sorting through 1000 glp_poses. The AI mode of google was used to help create scripts in Python which interface with PyMol [5]. The scripts were run through a laptop terminal (MacOs) to enable batch processing without crashing PyMol. The scripts created can be found in the github link listed above. The logic of the scripts created filtered for glp_poses which interacted with GLP-1R(number of contacts was counted). The top result with the most contacts to xtal-lig was further analyzed in PyMol.

III. Results

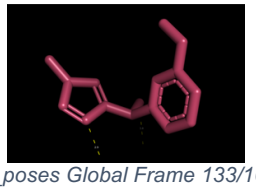

Figure 1: glp_poses Global Frame 133/1000 (Agonist)

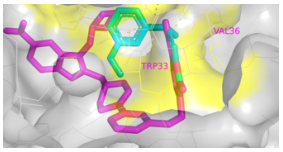

Figure 2: Binding Pocket of GLP-1R to xtal-lig +: glp_poses Global Frame 133/1000 (Agonist)

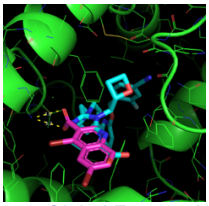

Figure 3: glp_poses Global Frame 734/1000 (Agonist)

IV. Discussion and Conclusion

The top result for agonists of GLP-1R in the active state was glp_poses 133. It has high affinity to GLP-1R contacting it with a total of 27 atoms. It sits in the same binding pocket with a distance of 0.82 Å of xtal-lig and is 1.05Å deeper than xtal-lig. It has interactions with TRP33 and VAL36. It has hydrogen bonds and does not sterically clash with GLP-1R. The glp_poses 154,659,386, and 411 can be evaluated visually in the future for further analysis of competitive agonists in GLP-1R activation. The worst result was 734. It does not interact with GLP1-R.

REFERENCES

[1] Zheng, Z., Zong, Y., Ma, Y. *et al.* "Glucagon-like peptide-1 receptor: mechanisms and advances in therapy", *Sig Transduct Target Ther* **9**, 234 (2024). https://doi.org/10.1038/s41392-024-01931-z

[2] Yuan, S., Xia, L., Wang, C., Wu, F., Zhang, B., Pan, C., Fan, Z., Lei, X., Stevens, R. C., Sali, A., Sun, L., & Shui, W. (2023) "Conformational Dynamics of the Activated GLP-1 Receptor-G_s Complex Revealed by Cross-Linking Mass Spectrometry and Integrative Structure Modeling", *ACS central science*, *9*(5), 992–1007. https://doi.org/10.1021/acscentsci.3c00063

[3] Eric Lewandowski, Ph.D., "Computational Modeling and Agonists Discovery," presented at BCH6746 University of South Florida, April 14,2026 [PowerPoint Slide 2]

[4] PyMOL
The PyMOL Molecular Graphics System,
Version 3.1 Schrödinger, LLC.

[5] The code in https://github.com/jmmoses99/Structural_Bio.git was generated using Google Gemini 2.5 Pro (April 2026)

No, generative artificial intelligence (AI) or other large language model (LLM) tool, such as ChatGPT, was used to prepare this document.